

Question	Answer	Marks
4(a)	- molecules are in (constant) random motion (all) collisions between molecules are (perfectly) elastic - no forces between molecules (except during collisions) - volume of molecules is negligible (compared with volume of gas) - collisions involving molecules are instantaneous <i>Any three points, 1 mark each</i>	B3
4(b)	- molecules collide with (walls of) container - momentum of molecule changes during collision (with walls) - change in momentum is caused by force on molecule by wall - molecule experiences force from wall so molecule exerts force on wall - many molecules exerting force across the area of the wall leads to pressure (on the wall) <i>Any three points, 1 mark each</i>	B3
4(c)	<i>Any three bulleted points from:</i> - both gases are ideal <i>Up to 2 points from:</i> - mass of one molecule of gas X is 3.3×10^{-27} kg - mass of one molecule of gas Y is 6.6×10^{-27} kg - mass of one molecule of gas Y is double mass of one molecule of gas X <i>Up to 2 points from:</i> - sample of X contains 0.27 mol / 1.6×10^{23} molecules - sample of Y contains 0.81 mol / 4.9×10^{23} molecules - sample of Y contains treble the amount of gas / number of molecules as sample of X <i>Up to 2 points from:</i> - mass of gas X is 5.4×10^{-4} kg - mass of gas Y is 3.2×10^{-3} kg - mass of gas Y is six times mass of gas X	B3

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